

Crime Analysis

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Contents

1	Crime Incidents Reports Data	1
2	Explorative Analysis	1
2.1	Incidents by Day of the Week	1
2.2	Year-over-Year Analysis	4

1 Crime Incidents Reports Data

```
path <- here::here("SoSe_2026/Rmarkdown/Exercise/")
crime_data <- readRDS(file.path(path, "crime_data.RDS"))
crime_desc <- readxl::read_xlsx(file.path(path, "FieldExplanation.xlsx"))
```

```
head(crime_data) %>% gt()
```

The dataset contains 261864 incident reports described using 17 variables.

```
crime_desc %>%
  separate(col = `Field Name, Data Type, Required`,
            into = str_split(names(crime_desc)[1],
                              pattern = ",")[[1]], sep = ",") %>%
  mutate("Field Name" = str_replace_all(`Field Name`,
                                          pattern = "[\\[\\]]",
                                          replacement = "")) %>%
  gt() %>%
  tab_caption("Test")
```

2 Explorative Analysis

2.1 Incidents by Day of the Week

INCIDENT_NUMBER	OFFENSE_CODE	OFFENSE_CODE_GROUP	OFFENSE_DESCRIPTION
232007173	3115	NA	INVESTIGATE PERSON
232004454	3301	NA	VERBAL DISPUTE
232006290	3115	NA	INVESTIGATE PERSON
232024939	3114	NA	INVESTIGATE PROPER
232006708	423	NA	ASSAULT - AGGRAVAT
232007629	3115	NA	INVESTIGATE PERSON

Table 1: Test

Field Name	Data Type	Required	Description
incident_num	[varchar](20)	NOT NULL	Internal BPD report number
offense_code	[varchar](25)	NULL	Numerical code of offense description
Offense_Code_Group_Description	[varchar](80)	NULL	Internal categorization of [offense_de
Offense_Description	[varchar](80)	NULL	Primary descriptor of incident
district	[varchar](10)	NULL	What district the crime was reported
reporting_area	[varchar](10)	NULL	RA number associated with the wher
shooting	[char] (1)	NULL	Indicated a shooting took place.
occurred_on	[datetime2](7)	NULL	Earliest date and time the incident c
UCR_Part	[varchar](25)	NULL	Universal Crime Reporting Part num
street	[varchar](50)	NULL	Street name the incident took place

```

weekdays_obj <- c("Sunday", "Monday", "Tuesday",
                  "Wednesday", "Thursday", "Friday", "Saturday")
weekly_stats <- crime_data %>%
  mutate(DAY_OF_WEEK = factor(DAY_OF_WEEK,
                              levels = weekdays_obj)) %>%
  group_by(DAY_OF_WEEK) %>%
  summarise(incident_count = n()) %>%
  mutate(is_max = incident_count == max(incident_count))

```

```

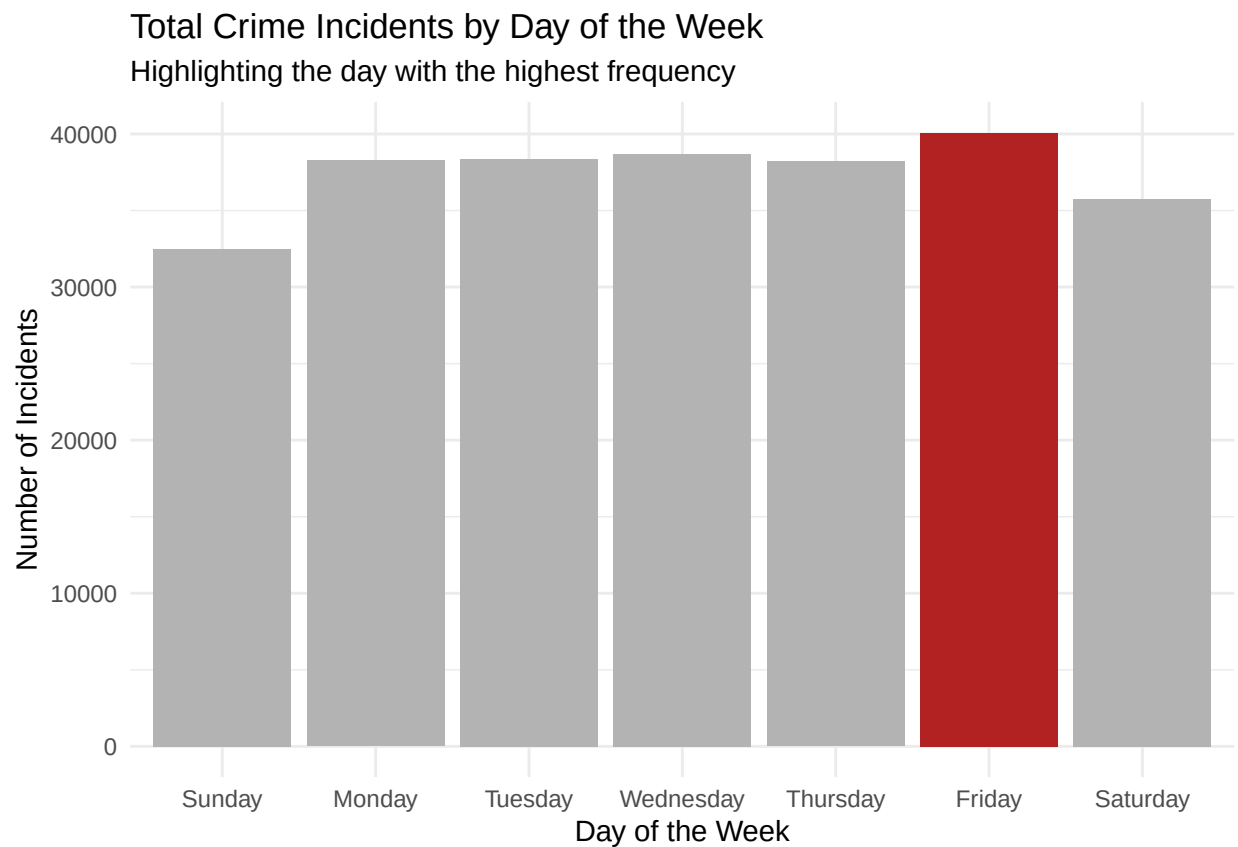
ggplot(weekly_stats, aes(x = DAY_OF_WEEK, y = incident_count, fill = is_max)) +
  geom_col() +
  scale_fill_manual(values = c("gray70", "firebrick"), guide = "none") +
  labs(title = "Total Crime Incidents by Day of the Week",
       subtitle = "Highlighting the day with the highest frequency",
       x = "Day of the Week",
       y = "Number of Incidents") +
  theme_minimal()

```

Table 2: The data shows that Friday typically records the highest volume of incidents.

Weekly Crime Distribution

Day	Total Reports
Sunday	32482
Monday	38263
Tuesday	38357
Wednesday	38713
Thursday	38208
Friday	40085
Saturday	35756



```
weekly_stats %>%
  select(DAY_OF_WEEK, incident_count) %>%
  gt() %>%
  tab_header(title = "Weekly Crime Distribution") %>%
  tab_caption(caption = "The data shows that Friday typically records
    the highest volume of incidents.") %>%
  cols_label(DAY_OF_WEEK = "Day", incident_count = "Total Reports")
```

2.2 Year-over-Year Analysis

```
incidents_by_month <- crime_data %>%  
  summarise(Incidents = n(), .by = c(YEAR, MONTH)) %>%  
  mutate(Month = make_date(YEAR, MONTH, 1))  
  
gg <- incidents_by_month %>%  
  ggplot(aes(x = Month, y = Incidents)) +  
    geom_line(col = "darkblue") +  
    geom_point(col = "darkblue") +  
    labs(title = "Incidents by Month") +  
    theme_minimal()  
  
if (knitr::is_html_output()) {  
  plotly::ggplotly(gg)  
} else if (knitr::is_latex_output()) {  
  print(gg)  
}
```

